IBM Cloud Pak for Business Automation Demos and Labs 2026

Use an AI Agent to Validate Banking Information

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1 Lab Introduction

This hands-on lab explores IBM Business Automation Workflow (BAW) enhanced by AI Agents deployed in IBM watsonx Orchestrate (wxO) to streamline business processes and improve decision-making. Participants will gain practical experience integrating AI Agent capabilities into workflows.

1.1 Lab Scenario

Banking validation based on customer-supplied bank statements is one of the activities in the Review Client Documents steps (step with the green border below).

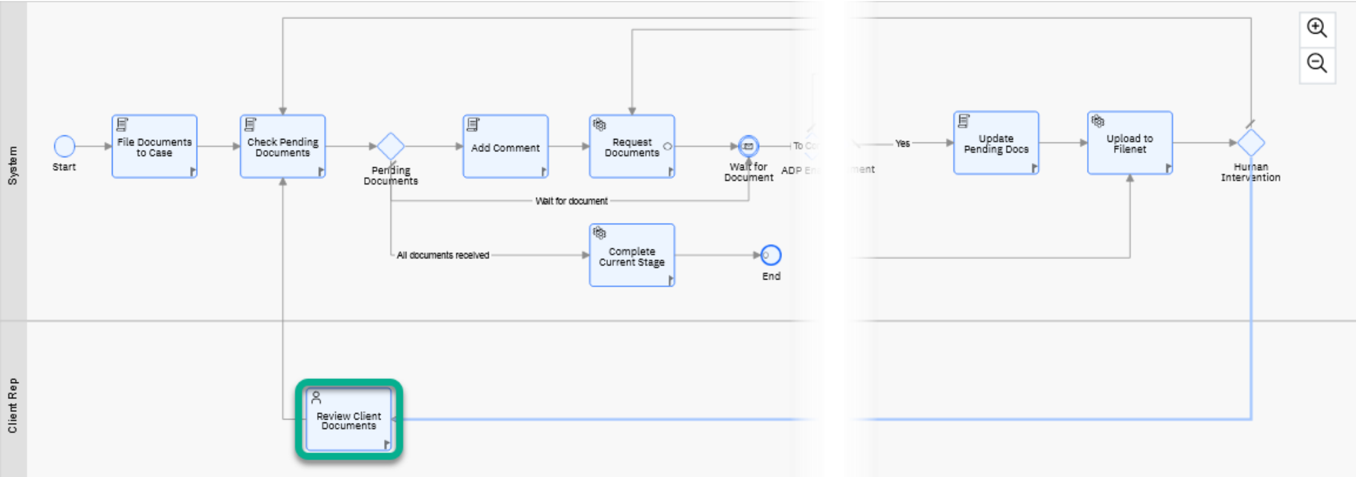
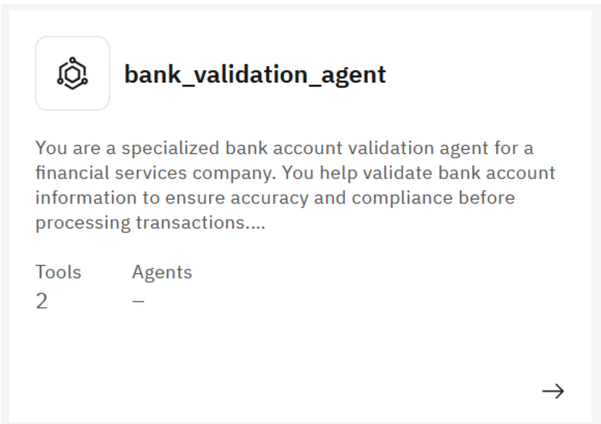


Figure 1. Client Onboarding Case: Review Documents Process

In this lab, you will create an automation that uses an AI Agent to help the client onboarding specialist validate the client's banking information.

1.2 AI Agent

The **bank_validation_agent** AI Agent you will be invoking to validate the client's banking information is deployed in an IBM watsonx Orchestrate tenant.



Typically, an AI Agent is invoked via a Chat interface, as shown in the figure below. In your lab, you will use the AI Agent Service in IBM Business Automation Workflow to interact with the AI Agent as an API.

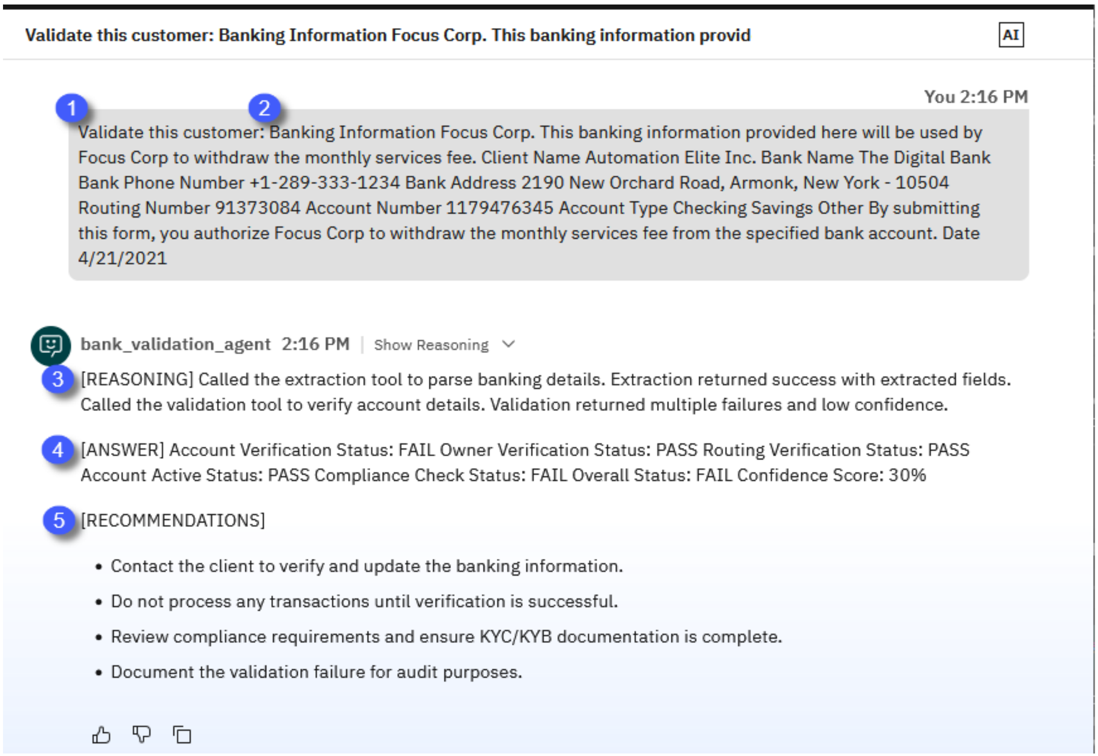


Figure 2. Banking Validation AI Agent

The typical flow of invoking the agent via the chat would be:

1. The trigger phrase for the AI Agent to start the validation process.
2. The bank account information was extracted from the client's Bank Statement.
3. Since this is a React Agent, we instructed the AI Agent to display the reasoning: all the tools it called and the outcome.
4. The final answer shows the status of all the checks performed by the AI Agent.
5. The AI Agent displays the validation-related recommendations.

1.3 Lab Setup Instructions

If you are performing this lab as a part of an **IBM event:**

- Access the document that lists the available systems and URLs along with login instructions. For this lab, you will need to access **IBM Business Automation Studio**.
- Reach out to one of your instructors to get the required **IBM watsonx Orchestrate API key**.

If you are performing this lab as a **self-paced** lab in a Client Onboarding enabled environment:

Find the URL for **IBM Business Automation Studio**, **user information** and the **IBM watsonx Orchestrate API key** in the **ConfigMap "000-client-onboarding-information"** of the IBM Cloud Pack for Automation project in the OpenShift Web Console.

In any case **download** the two banking information PDFs from the **Lab Data** folder in GitHub to a local folder of your choice.

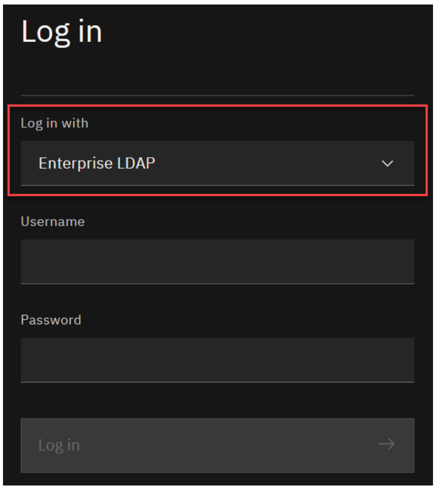
Banking Information - Automation Elite Inc.pdf

Legacy Consulting - Banking Information.pdf

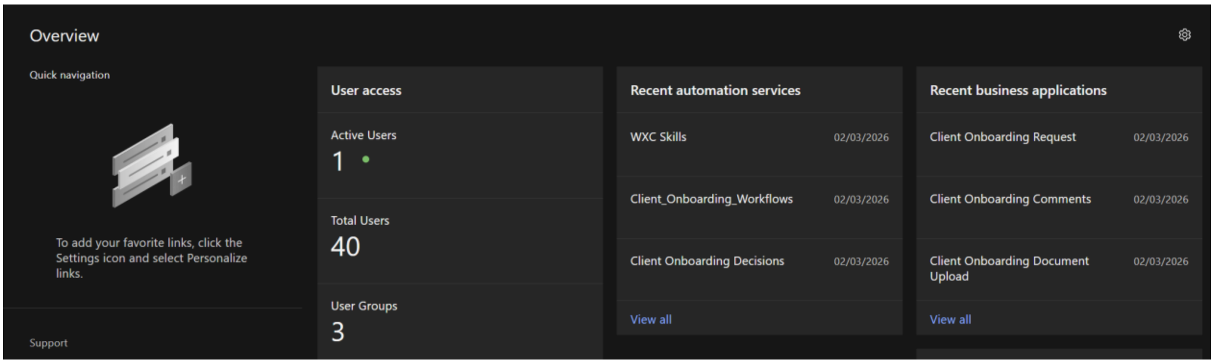
2 Lab Instructions

2.1 Preparation

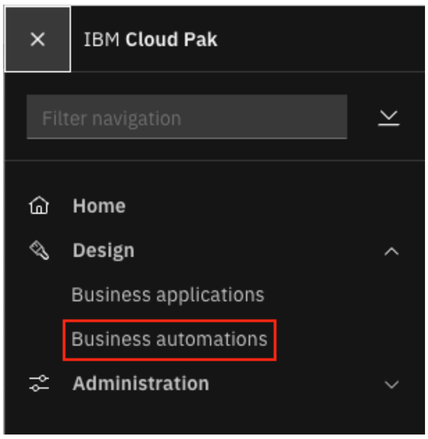
_1. In your browser, **login** to **IBM Business Automation Studio** using the **Enterprise LDAP** option.



The homepage contains cards that showcase recent artifacts across all installed Cloud Paks in the system. For IBM Cloud Pak for Business Automation, the recent business applications and automation services are shown.

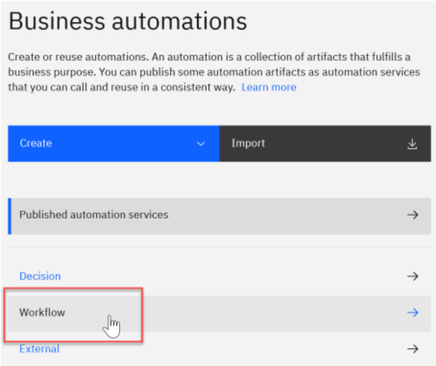


_2. In the top-left corner, **click** on the **Hamburger menu** icon and select **Design → Business automations** to access the automation repository.



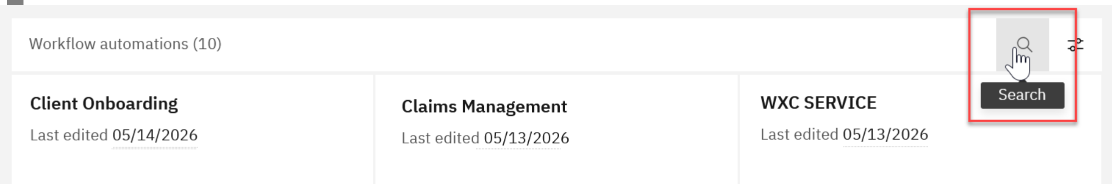
This brings up the Business automations page where you can create or reuse automations from different capabilities of IBM Cloud Pak for Business Automation. If a capability is not installed in the environment, it will be greyed out.

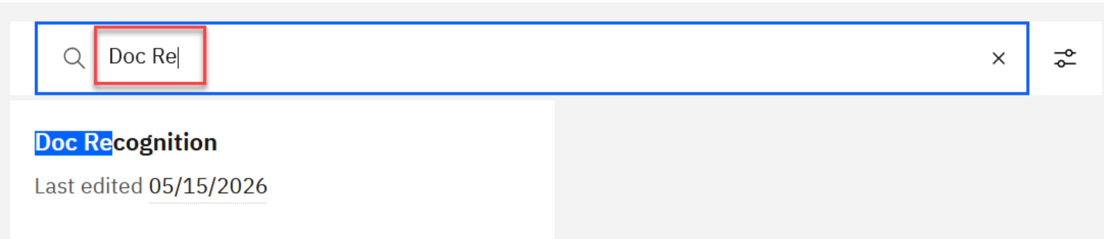
_3. **Click** on **Workflow** to display all available Workflow solutions.



_4. **Find** the Workflow solution **Doc Recognition** or **click** the **magnifying glass** icon in the top right corner and enter **Doc Re** to filter all Workflow Solutions down to only this solution.

> In a multi-user environment you might find other solutions containing Doc Recognition, for example prefixed with UsrXXX.



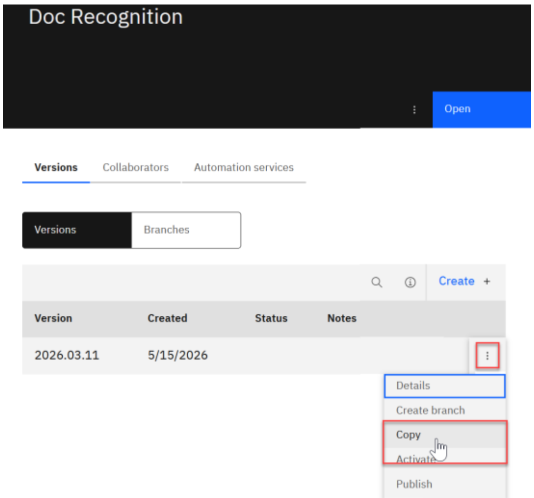


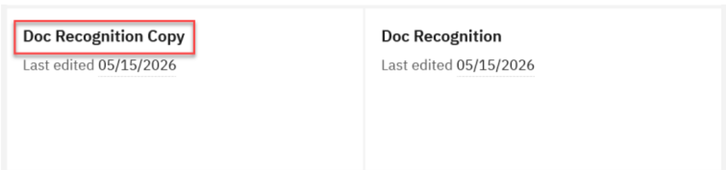
> In case you have previously done the lab "Use Gen AI to Identify and Validate Documents", you can skip steps _5 - _8 and directly continue with step _9.

_5. **Click** on the **tile** of the **Doc Recognition** solution but not the Open link.

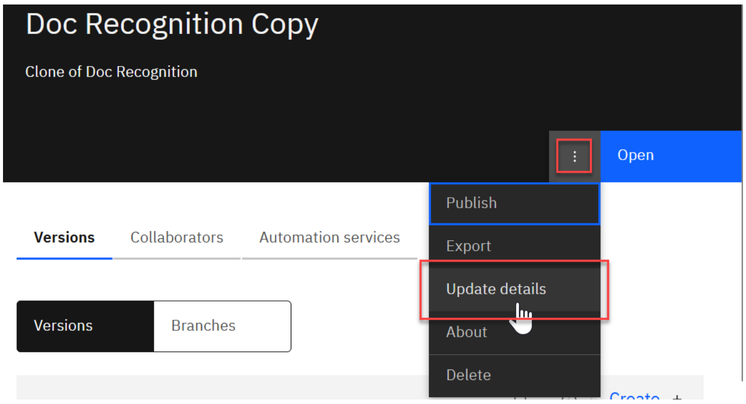
_6. **Click** on the **three vertical docs** at the right end of the **single version** that exists (the version number may differ from the one in the screenshot) and **click Copy**.

This will create a copy of the Doc Recognition solution with Copy appended to the original name.



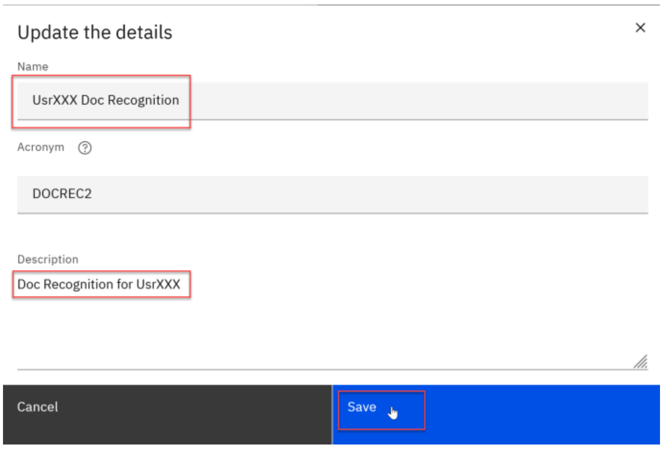


_7. **Click** on the **tile** (but not the open link), then **click** on the **three vertical dots** next to the blue Open button and **select Update details**.

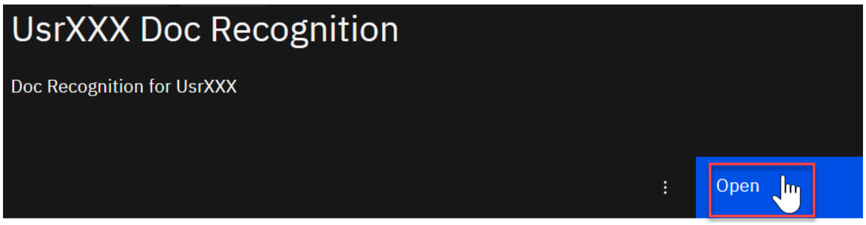


_8. **Modify** the **Name** of the solution to **UsrXXX Doc Recognition** and the **Description** to **Doc Recognition for UsrXXX** (replacing it with the name of the user you are using) before **clicking Save**.

> Keep the Acronym as is, independent of the value. This is auto generated to be unique and may differ from the screenshot.



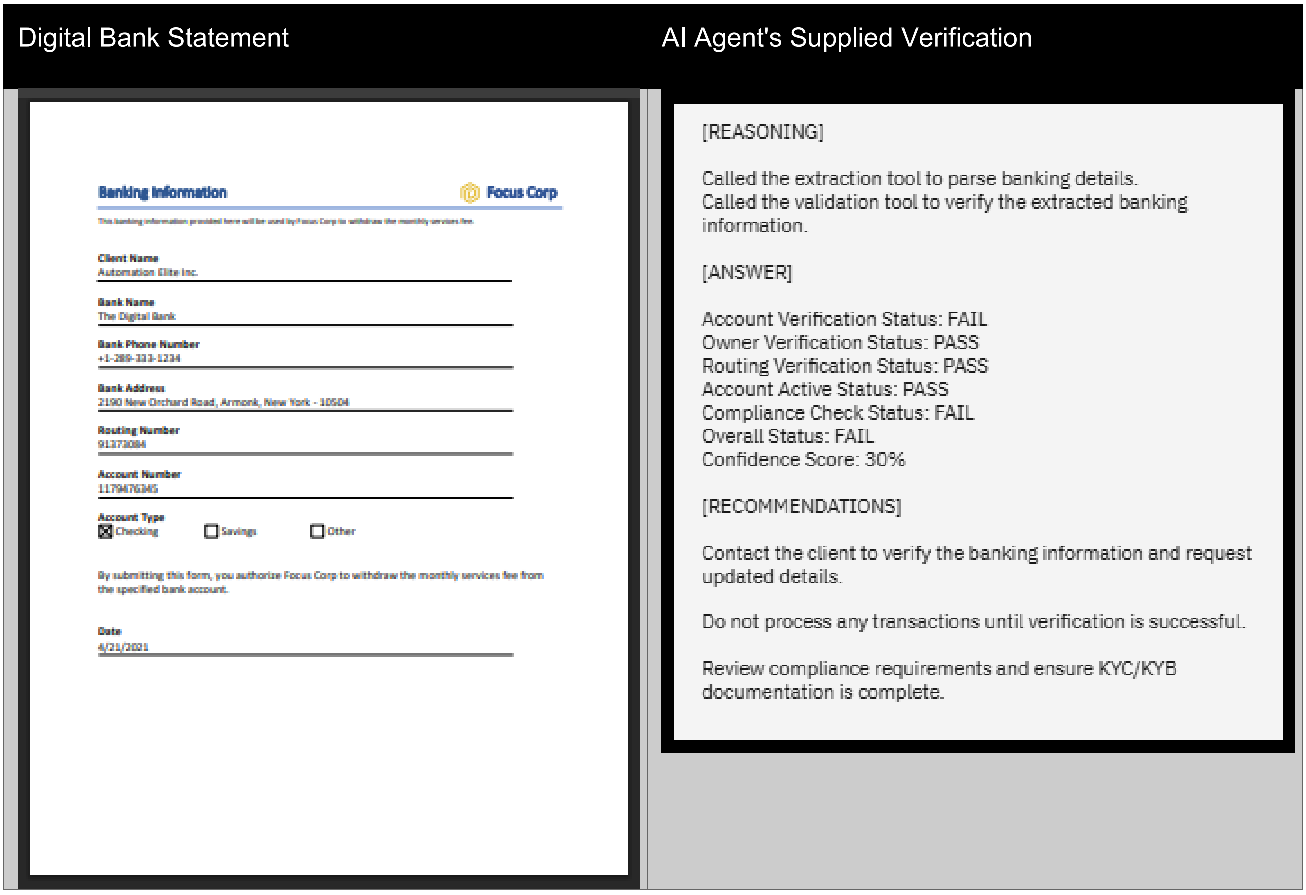
_9. **Click** on the blue **Open button** to open your *UsrXXX Doc Recognition* solution in Workflow Designer.



2.2 Author an AI Agent Service Flow

You will author an AI Agent activity to assist the client onboarding specialist in validating the client's banking status.

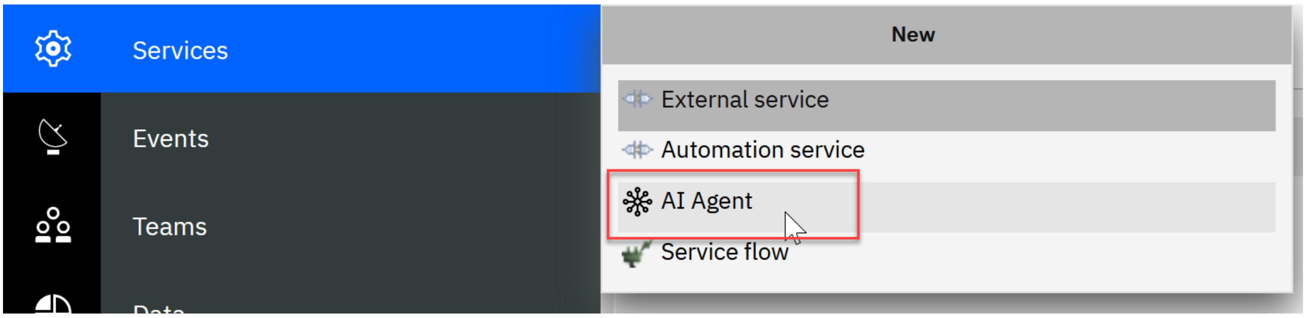
For example, given a banking statement, the AI Agent will determine the client's banking standing, as shown below.



2.2.1 Create an AI Agent Service

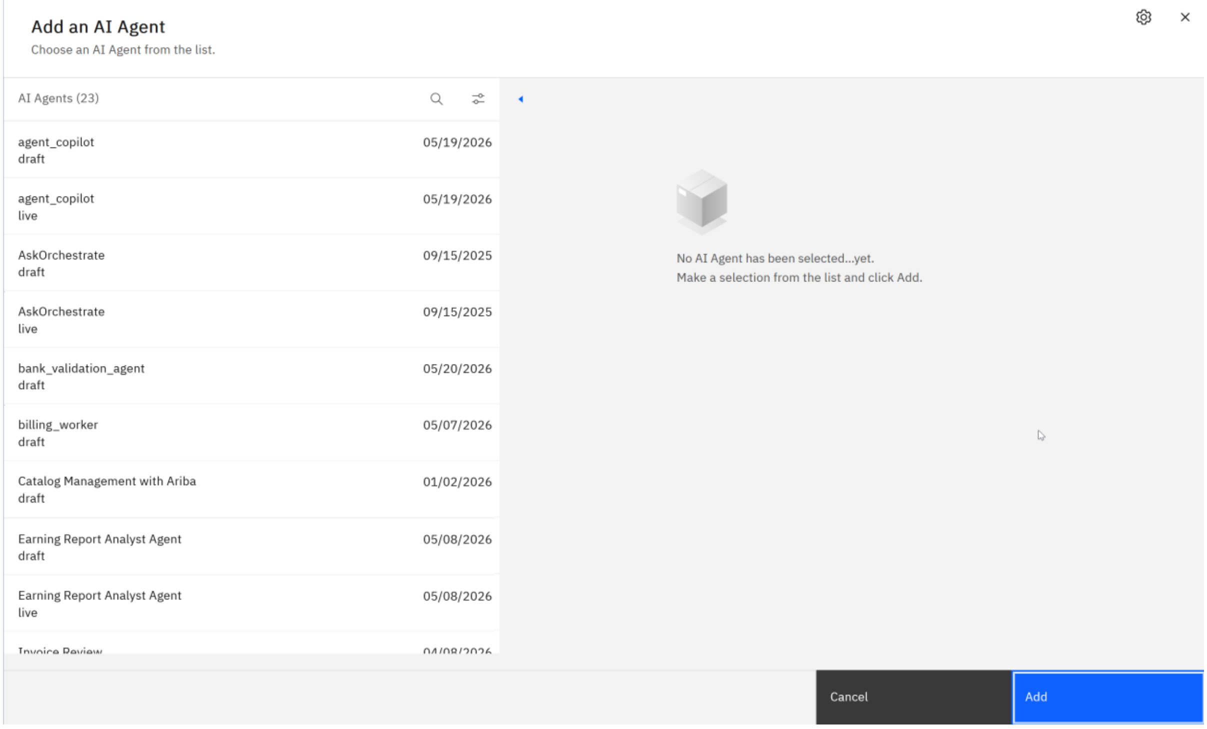
You will connect to the IBM watsonx Orchestrate tenant configured in your environment and find the Banking Verification Agent to create an AI Agent Service.

_10. From the palette, select **Services** > **+** > **AI Agent.**

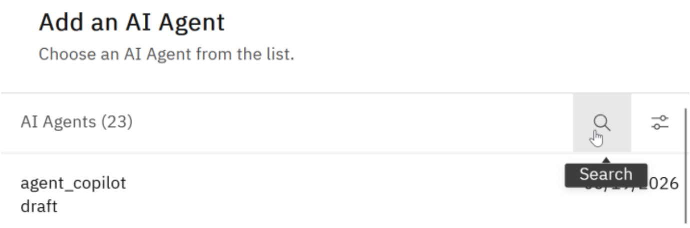


_11. **Enter** the **API Key** of the watsonx Orchestrate tenant the environment is connected to and **click Next** (refer to 1.3 Lab Setup Instructions for information how to obtain it).

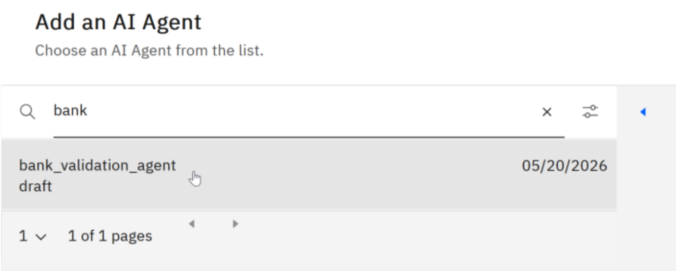
You should now see all AI Agents, both from the draft and the live environment, in the connected tenant. (The list of agents shown may be different in your environment.)



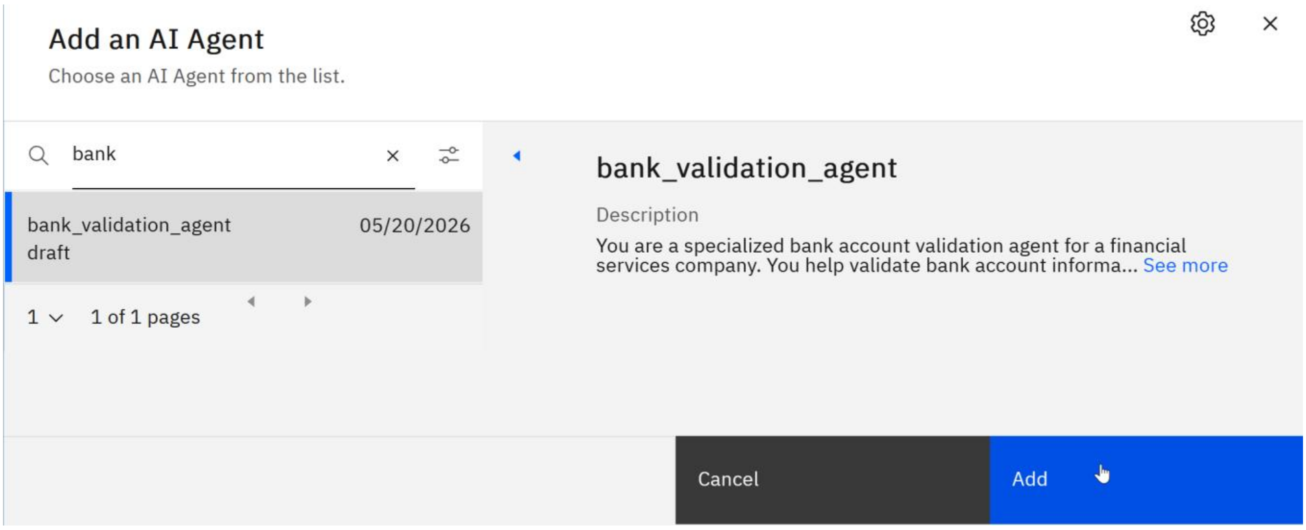
_12. **Click** the **Search** button.



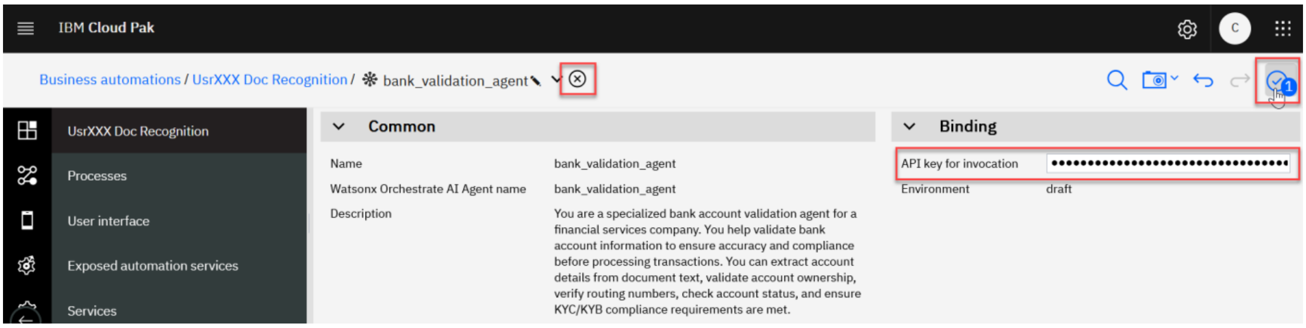
_13. **Type bank** and select **bank_validation_agent** AI agent in the **draft** environment.



_14. **Click** the **Add** button.



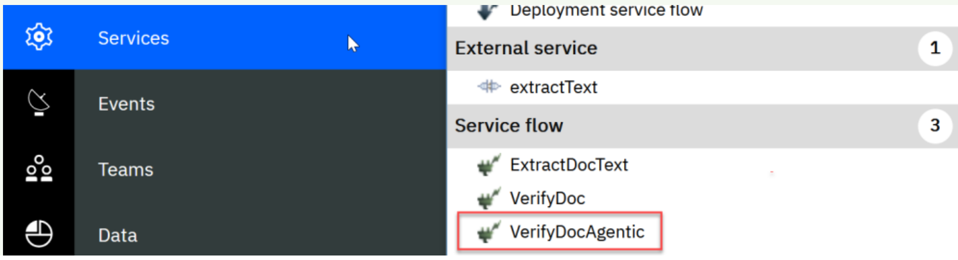
_15. In the **API key for invocation** field, **enter** the same **wxO API key** you used in step _11, **click Finish editing**, and **click x** to close the Service.



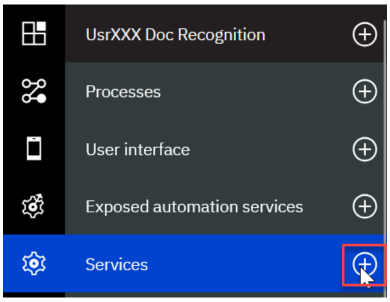
2.2.2 Create a Service Flow

You will create a Service flow that, given text extracted from a bank statement, verifies the client's banking information.

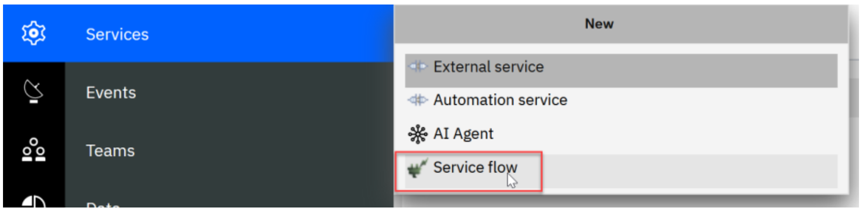
Note that we have already created a solution Service flow for you. You can use it as a reference.



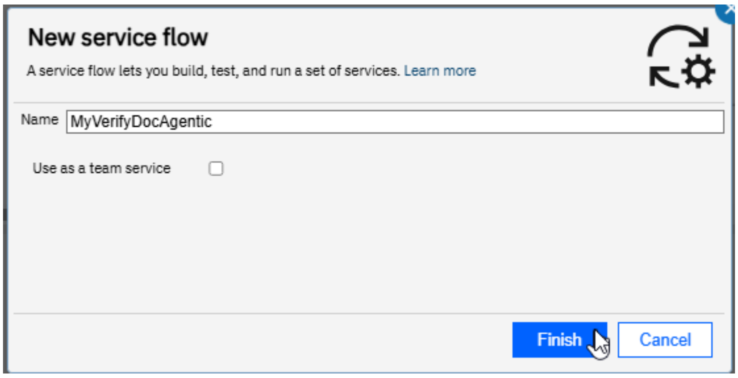
_16. **Click** the **+** on *Services.*



_17. **Click Service flow**.

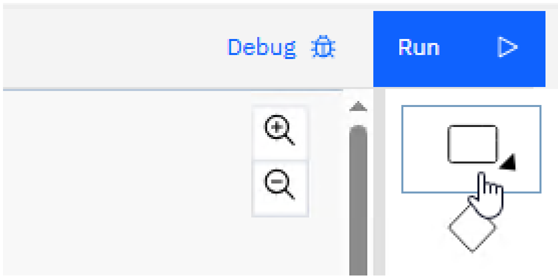


_18. **Enter MyVerifyDocAgentic** and **click Finish**.

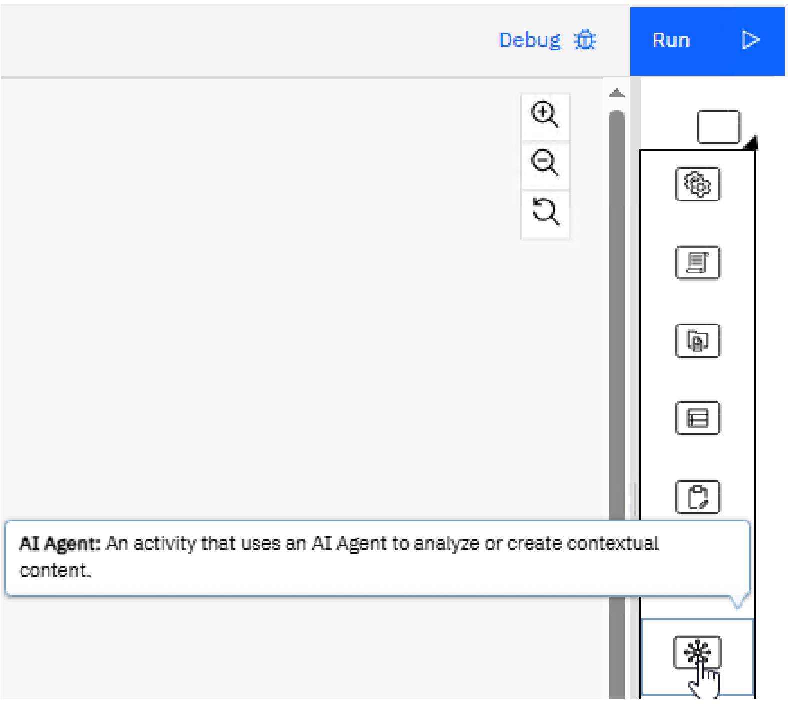


2.2.3 Add an AI Agent Activity

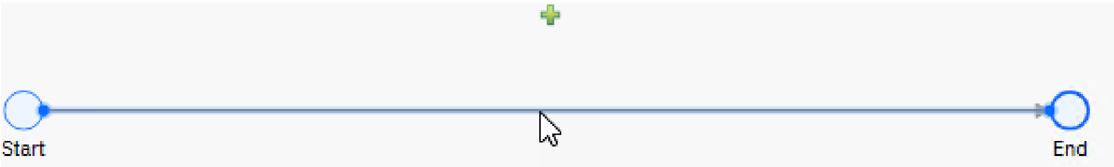
_19. **Click** the **Activity** icon in the palette.



_20. **Select** the **AI Agent** activity.



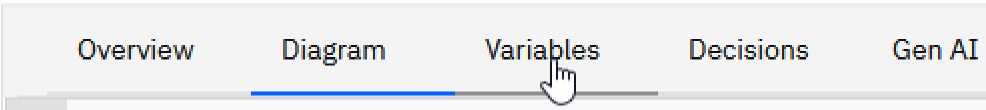
_21. While holding the left mouse button, **drag and drop** it onto the wire between the Start and End nodes.



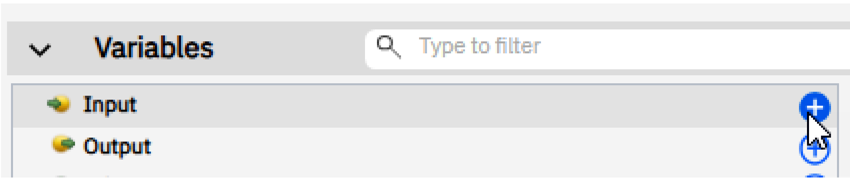
Ensure the AI Agent activity has been added as shown below:

2.2.4 Create Service Flow Variables

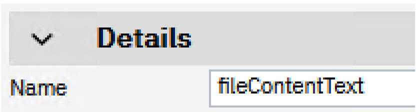
_22. **Click** the **Variables** tab.



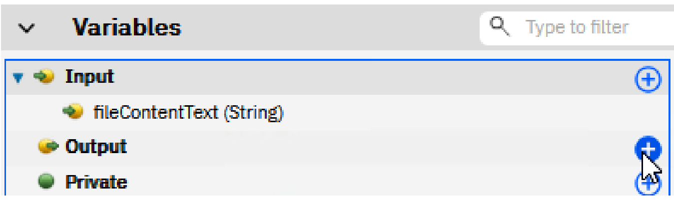
_23. **Click** the **+ icon** on the ***Input***.



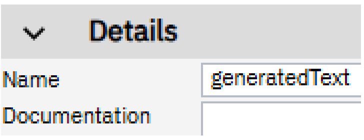
_24. For **Name**, enter **fileContentText.**



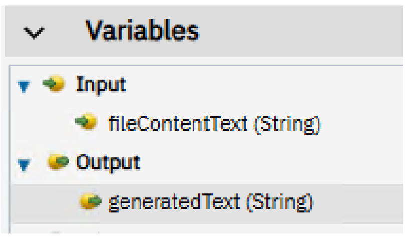
_25. **Click** the **+ icon** on the ***Output***.



_26. For **Name**, enter **generatedText.**

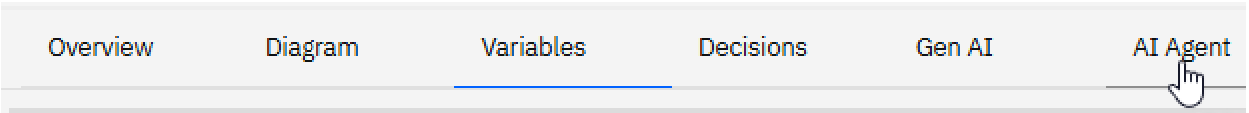


Verify that your variables match exactly what is shown below.

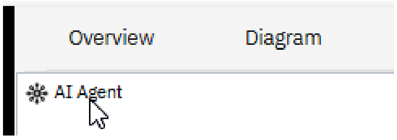


2.2.5 Author the AI Agent Activity

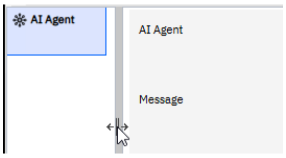
_27. **Click** the **AI Agent** tab.



_28. **Click** on **AI Agent.**



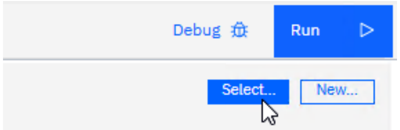
_29. **Select** the **divider** and **drag it to the left** to expand the AI Agent editor.



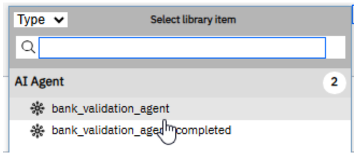
2.2.6 Select the AI Agent Service

You will now select the AI Agent Service you have just created.

_30. **Click** the **Select…** button.

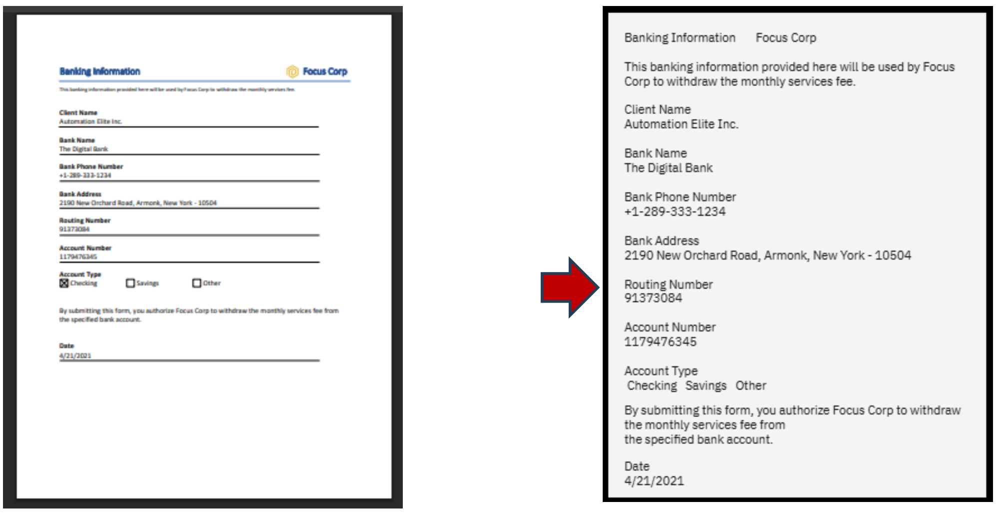


_31. Select the **bank_validation_agent**.



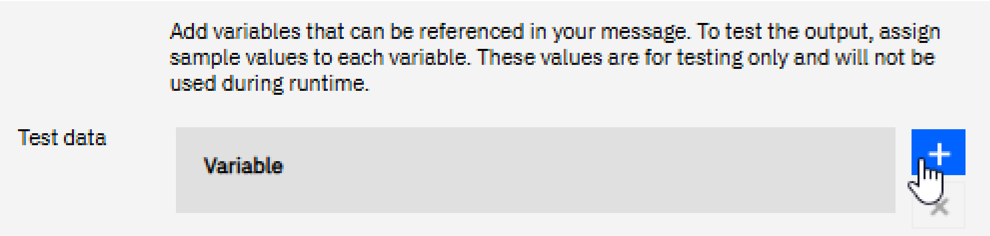
2.2.7 Create a Prompt Input Variable

The input variable will provide the AI Agent with the text extracted from the PDF bank statement file.

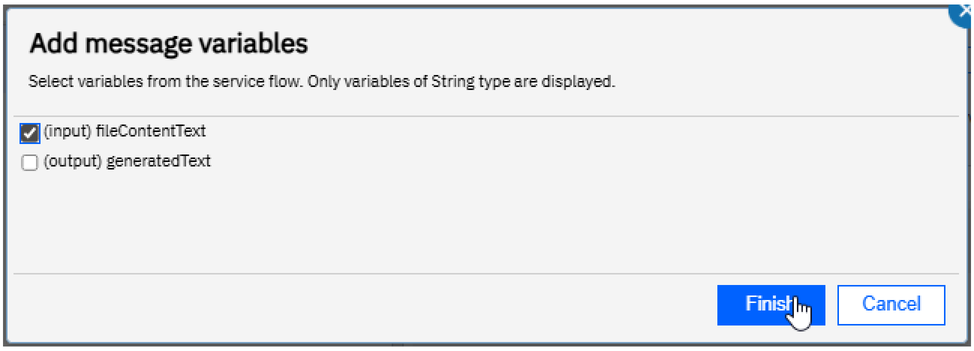


You will define the input the LLM will receive every time you call this Service flow.

_32. **Click** the **+** button to add an input variable.

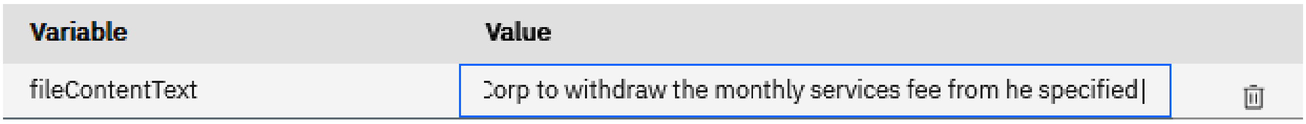


_33. **Select fileContentText** and **click Finish.**



_34. In the **Value column, enter** the **following text.**

```
Banking Information Focus Corp This banking information provided here will be used by
Focus Corp to withdraw the monthly services fee. Client Name Legacy Consulting Bank Name
XYZ Bank Bank Phone Number +1-213-111-7890 Bank Address 887 Cypress Rd, Garden Grove, CA
92840 Routing Number 32043559 Account Number 7250512345 Account Type Checking Savings
Other By submitting this form, you authorize Focus Corp to withdraw the monthly services
fee from the specified bank account. Date 5/10/21
```



Note: We initialized the variable so we can use it to test the AI Agent Service.

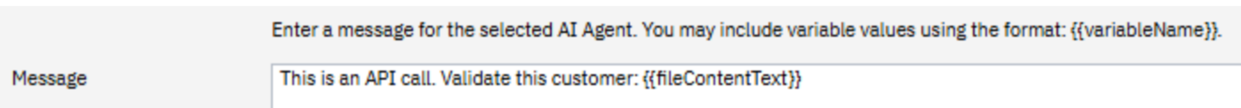
2.2.8 Define the Prompt Context

You will define a prompt to instruct the AI Agent to validate this customer.

_35. In the **Message** section, **add** the **text** below.

```
This is an API call. Validate this customer: {{fileContentText}}
```

The Message section should look exactly as shown below.



Note:

This is an API call – The Agent is instructed to format the output differently if the call comes from a source other than the Agent's chat window.

Validate this customer: – This is the command for the Agent to validate the data following the colon.

{{fileContentText}} – This is the actual data passed to the AI Agent from the Service flow's input variable.

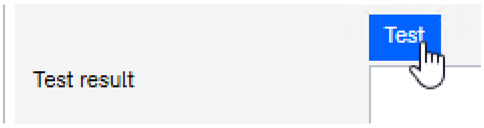
2.2.9 Test the AI Agent Task

We can use the default value for the input variable to test our AI Agent.

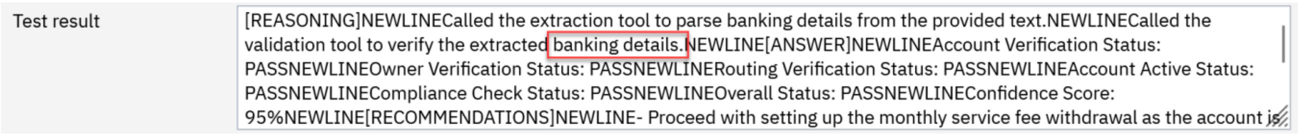
Variable	Value
fileContentText	ɔorp to withdraw the monthly services fee from he specified|

_36. **Click** the **Test** button.

> The invocation may take some time without a visual indication that the invocation is still running.



_37. **Verify** that the document is been handled for verification of the banking details.



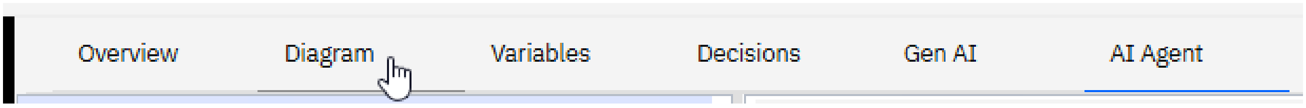
> **Note:** At this point, this is sufficient to confirm that the AI Agent integration worked. We will discuss the AI Agents' output message in more detail later in the lab.

2.2.10 Map the AI Agent Activity Output to Service Flow Variables

We will map the AI Agent output variable to the Service flow's output variable. This will allow the callers of the Service flow to interact with the AI Agent task.

> Note that we do not need to map the input variable explicitly because it has already been associated as input to the AI Agent in the AI Agent editor.

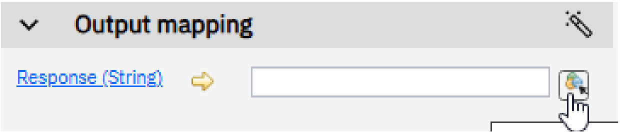
_38. **Click** the **Diagram** tab.



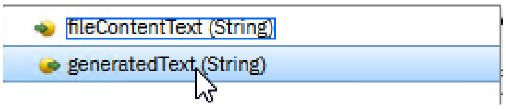
_39. **Select** the **AI Agent** activity.

_40. **Click** the **Data mapping** tab.

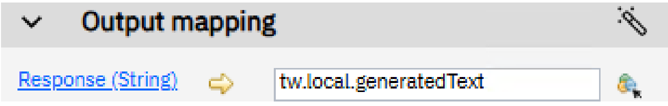
_41. On the ***Response (String)*** variable, **click** the **variable picker icon.**



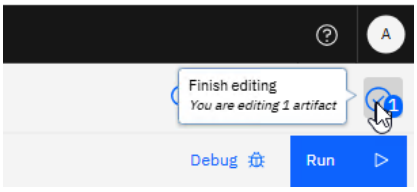
_42. **Select generatedText.**



Verify that the mapping looks as shown below.



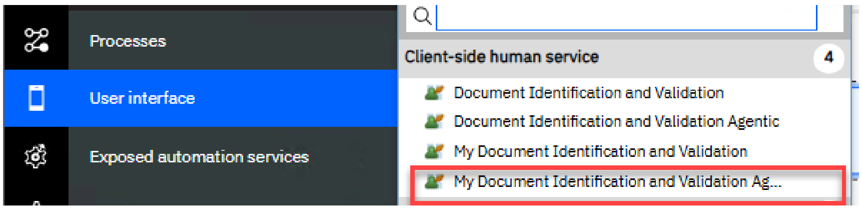
_43. **Click** the **Finish editing** icon to ensure that all your work is saved.



2.3 Add the AI Agent Service Flow to Human Task

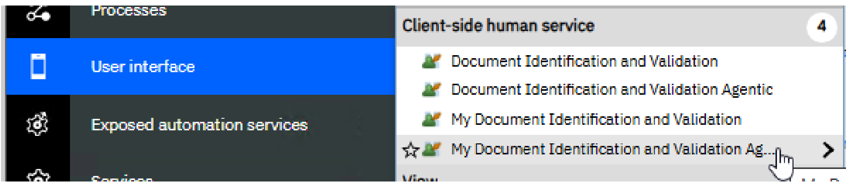
The AI Agent Service Flow (**MyVerifyDocAgentic**) you just authored is intended for use in the Human Task invoked from a process that supports a client onboarding specialist on validating the client's banking information.

In this lab, we have partially authored the Human Activity “*My Document Identification and Validation* Agentic“ (the official term is Client-side Human Service, or CSHS) for you. Your task is to add and configure the "MyVerifyDocAgentic" Service flow that you have just authored.

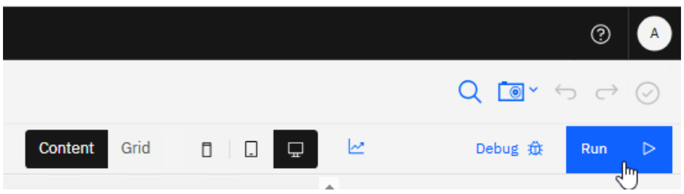


2.3.1 Examine the "My Document Identification and Validation Agentic" CSHS

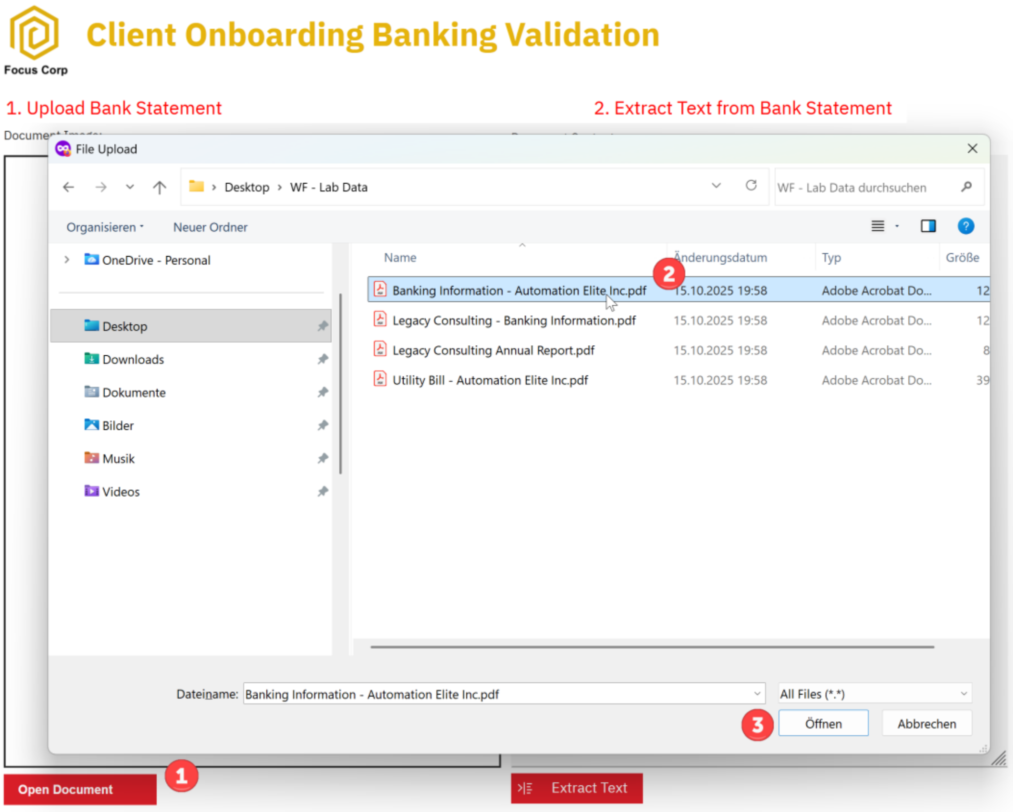
_44. **Click User interface,** and then **click My Document Identification and Validation Agentic.**



_45. **Click** the **Run** button (you may need to allow pop-ups).

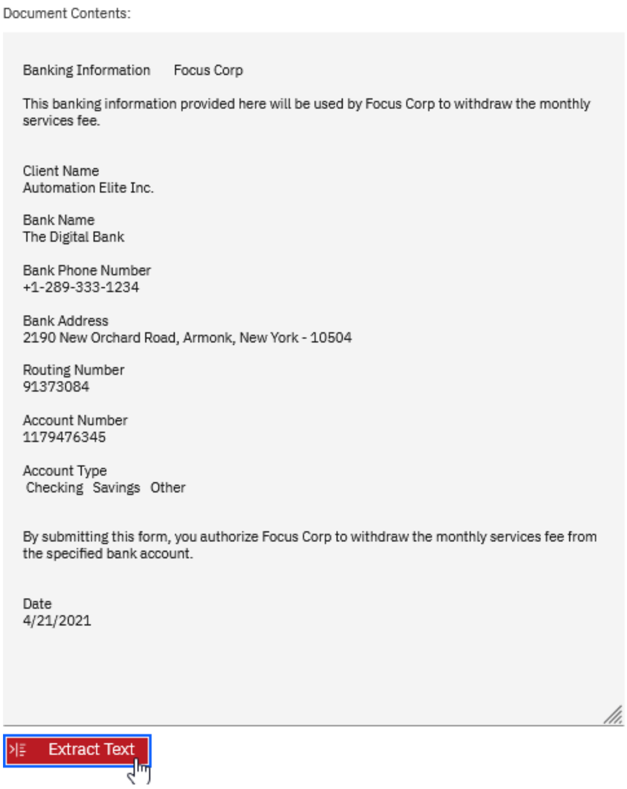


_46. **Click** the **Open Document** button, select **Banking Information – Automation Elite Inc.pdf**, in the folder you have saved the documents from the lab data to and **click Open.**



_47. **Click** the **Extract Text** button to extract text from the PDF document.

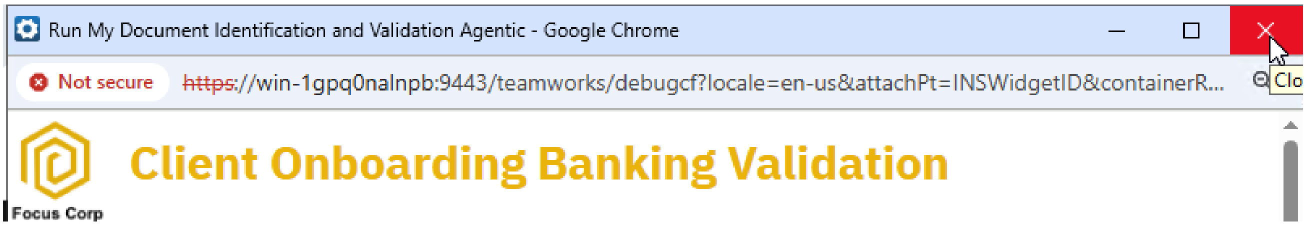
For this lab, a Java library is being used to extract the text from the PDF.



_48. **Click** the **Generate Summary** button and note that nothing happens.

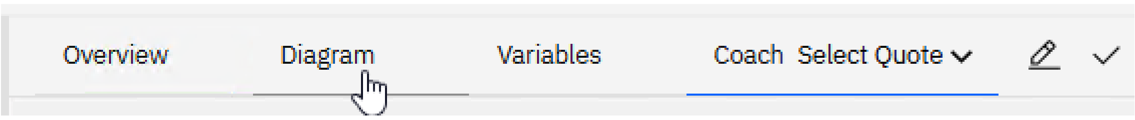
In the next few steps, you will configure the Client-side Human Service UI to send the document text to the AI Agent Service flow you created and display the output for the client banking validation.

_49. **Close** the browser **window or tab**.

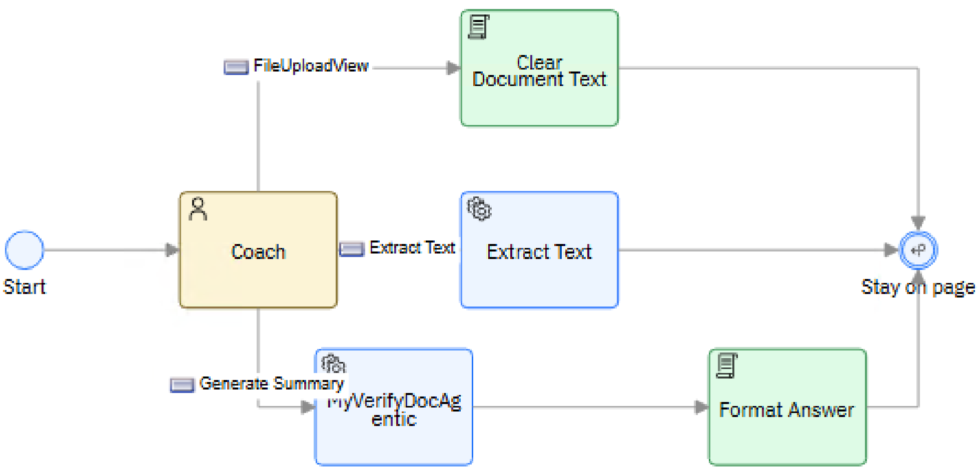


2.3.2 Add the "My MyVerifyDocAgentic" Service Flow

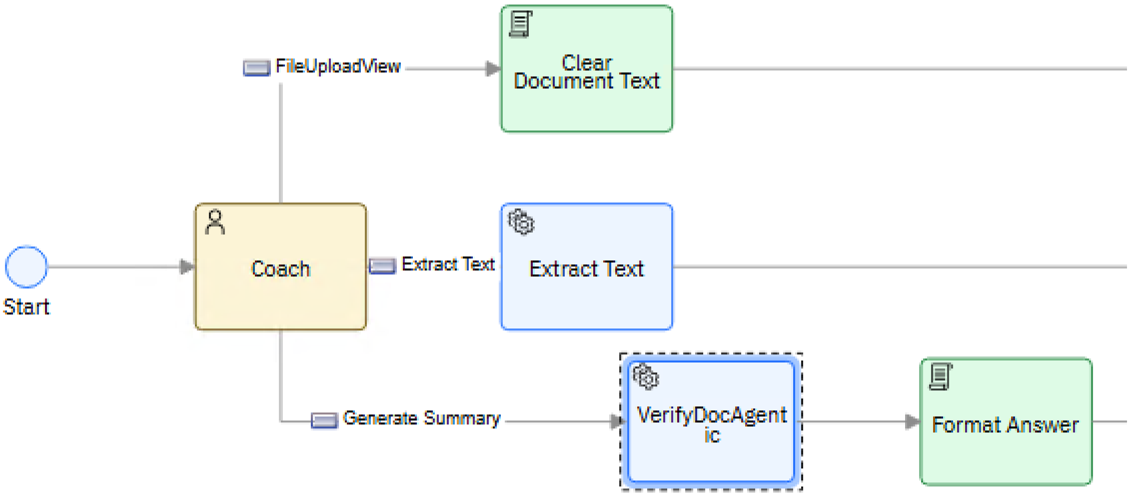
_50. **Click** the **Diagram** tab.



_51. **Click Services** on the left-hand side palette**,** then drag and drop the **MyVerifyDocAgentic** Service flow onto the **Generate Summary line** in the diagram.



Verify that the diagram now includes the MyVerifyDocAgentic Service flow you dragged, and it looks exactly as shown below.



2.3.3 Map the "MyVerifyDocAgentic" Service Flow Input/Output Variables

Before we perform variable mapping, let's examine the figure below to understand the flow and the variables involved.

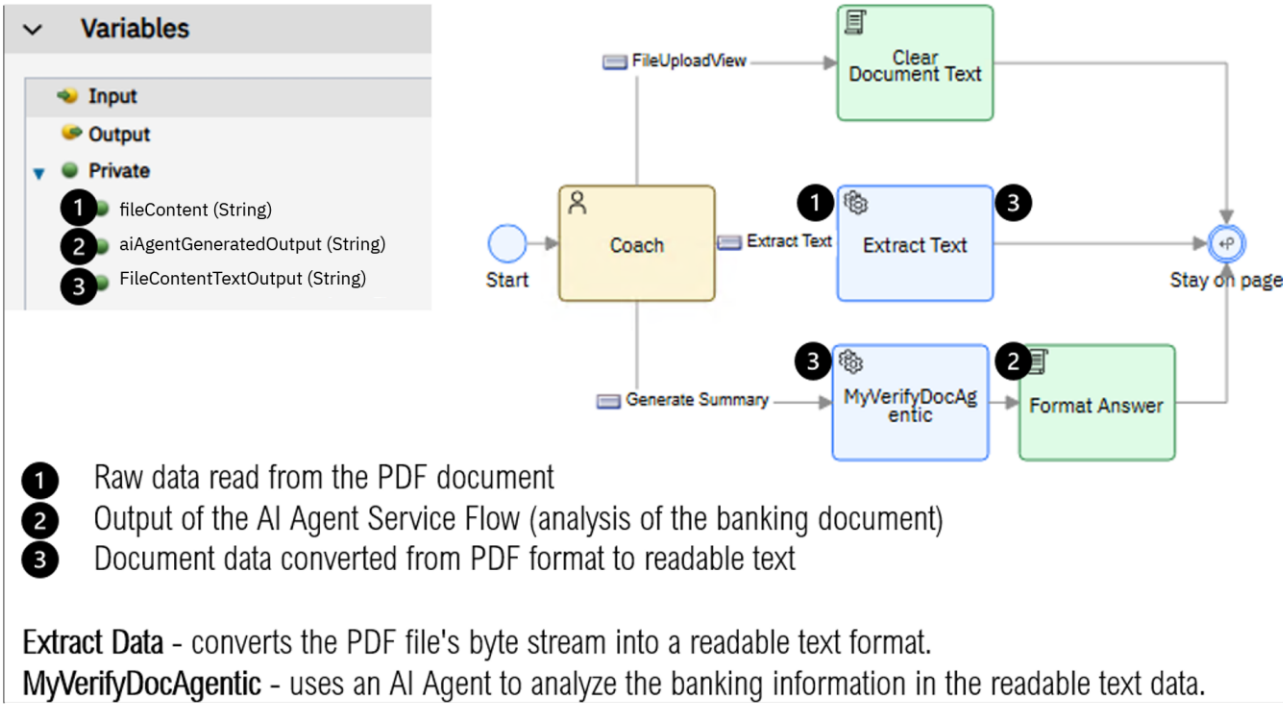
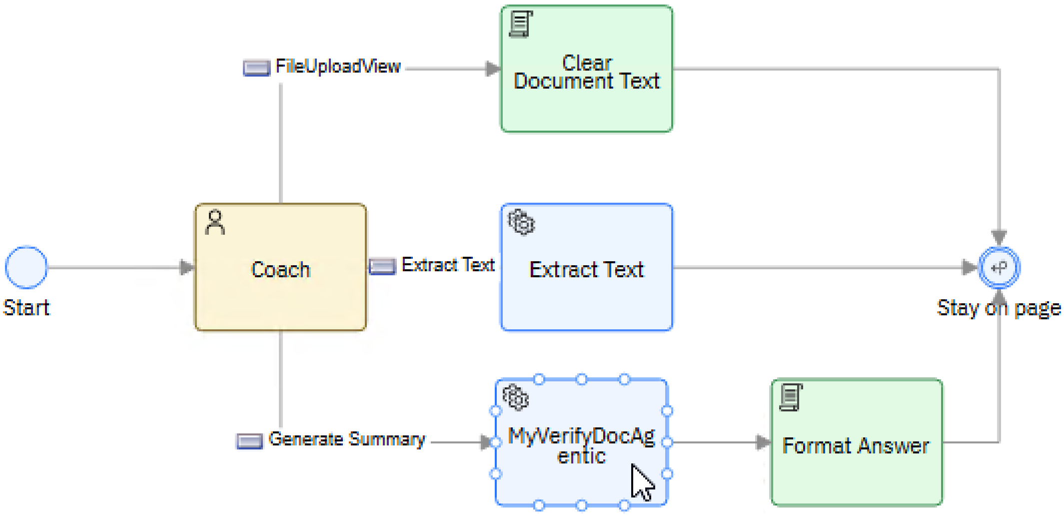
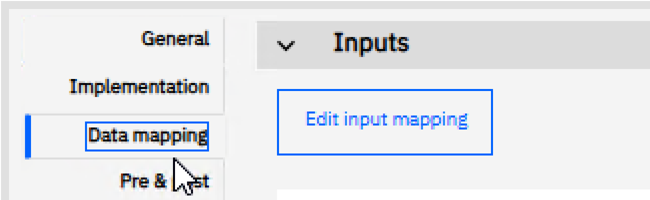


Figure 3. Service Flows and Variables

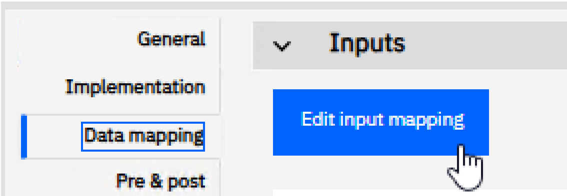
_52. **Click** the **MyVerifyDocAgentic** activity.



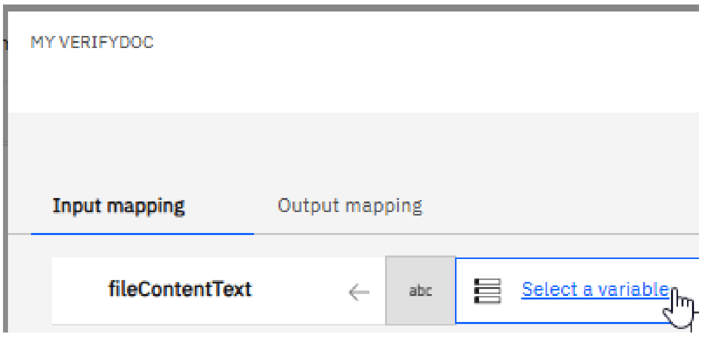
_53. **Click** the **Data mapping** tab.



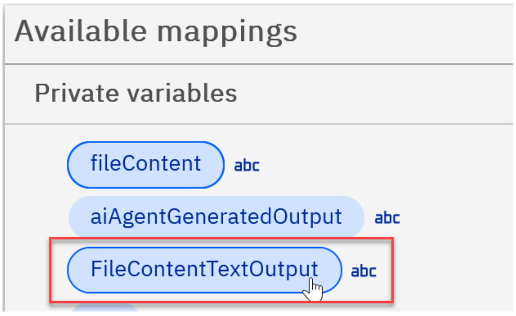
_54. **Click** the **Edit input mapping** button to open the Mapping Editor.



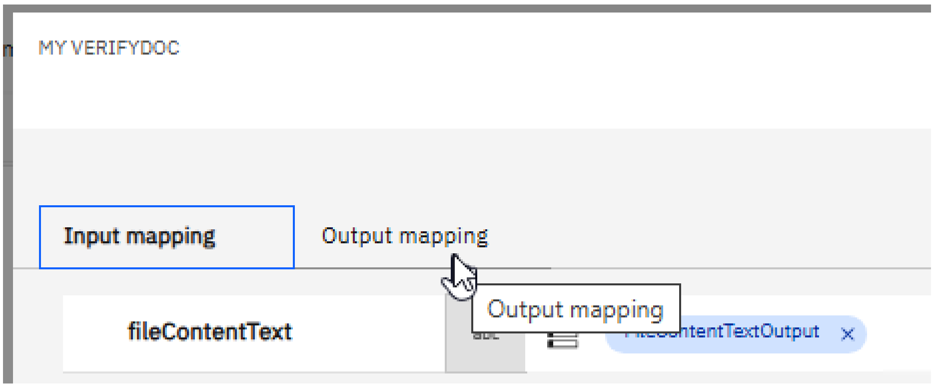
_55. **Click** the **Select a variable** link.



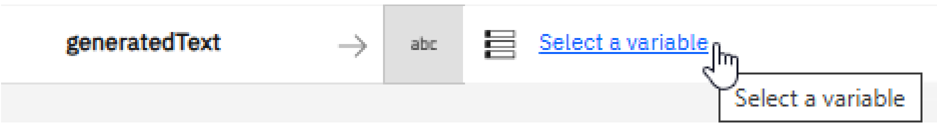
_56. **Select** the **FileContentTextOutput** variable.



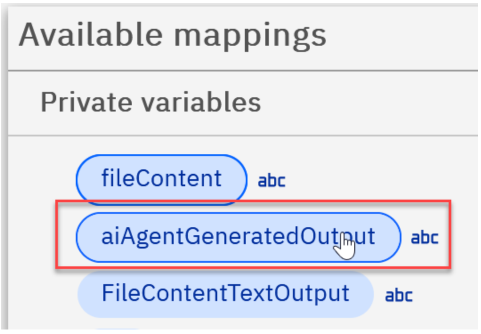
_57. **Click** the **Output mapping** tab.



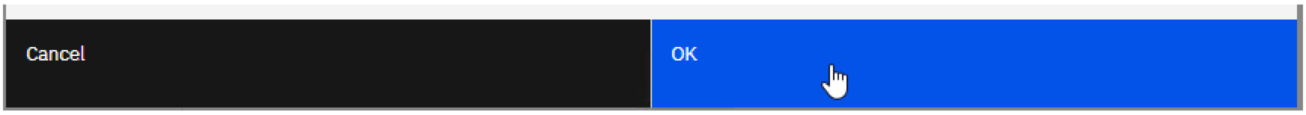
_58. **Click** the **Select a variable** link.



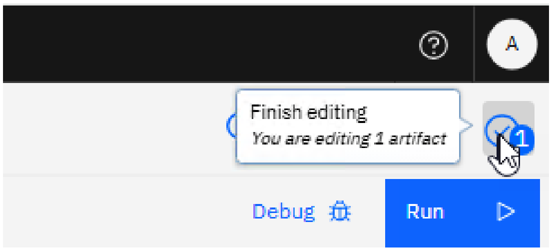
_59. Select the **aiAgentGeneratedOutput** variable.



_60. **Click OK** to close the Mapping Editor.

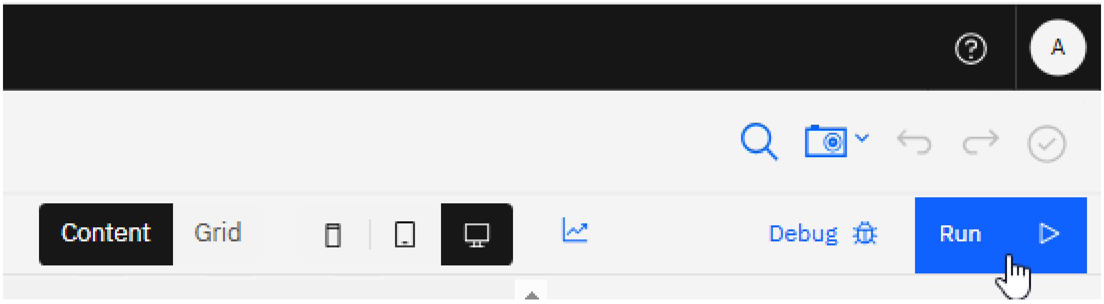


_61. **Click** the **Finish editing** icon to ensure that all your work is saved.



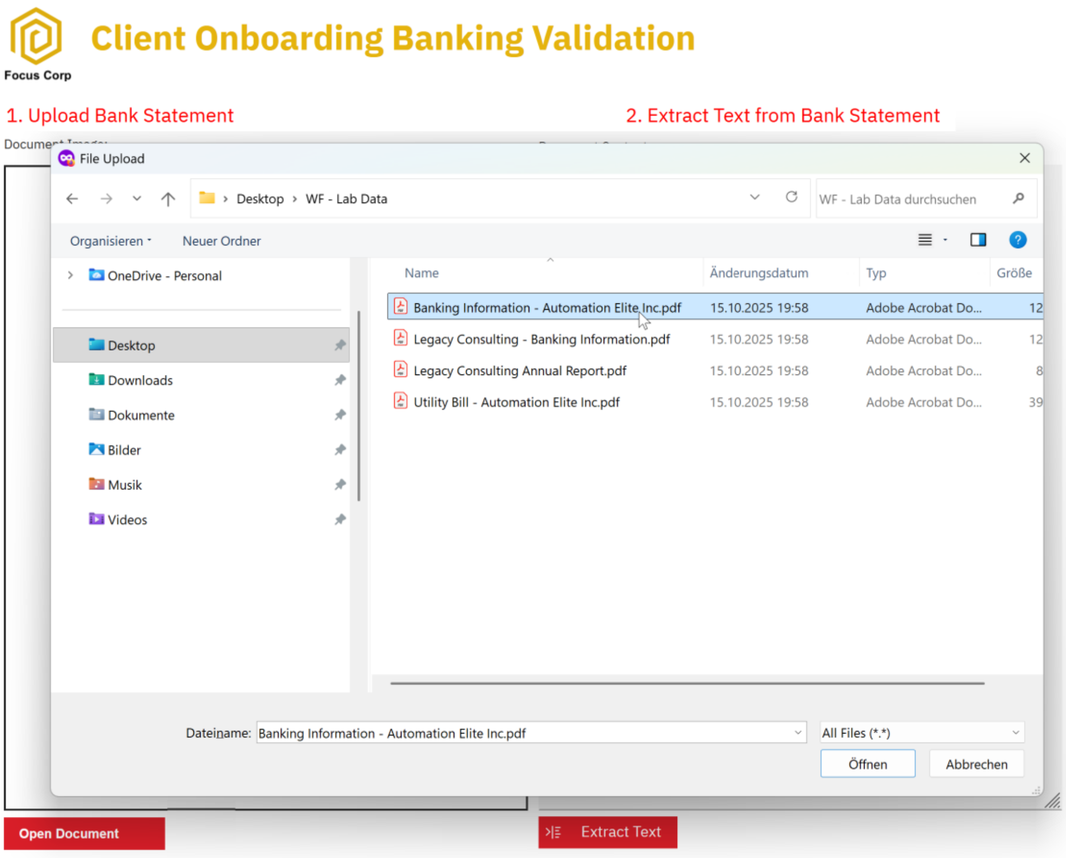
2.4 Test the "My Document Identification and Validation Agentic" CSHS

_62. **Click** the **Run** button.



_63. **Click** the **Open Document** button.

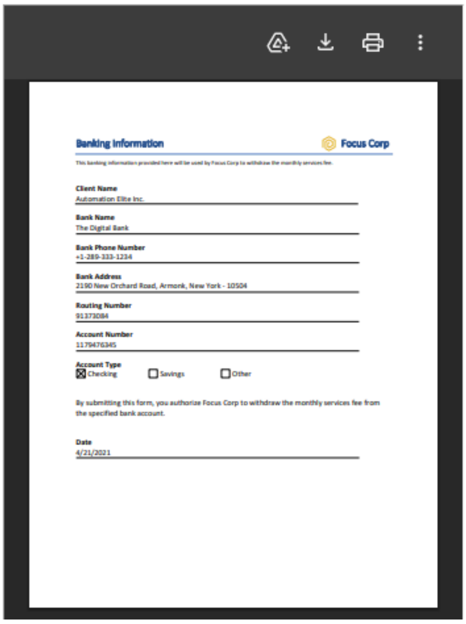
_64. **Select** again the **Banking Information - Automation Elite Inc.pdf** and **click Open.**



Verify that the document image appears as shown below:

1. Upload Bank Statement

Document Image:


Banking Information
Focus Corp

This banking information provided here will be used by Focus Corp to withdraw the monthly services fee.

Client Name
Automation Elite Inc.

Bank Name
The Digital Bank

Bank Phone Number
+1-289-333-1234

Bank Address
2190 New Orchard Road, Armonk, New York - 10504

Routing Number
91373084

Account Number
1179476345

Account Type
☒ Checking ☐ Savings ☐ Other

By submitting this form, you authorize Focus Corp to withdraw the monthly services fee from the specified bank account.

Date
4/21/2021

_65. **Click** the **Extract Text** button.

_66. **Click** the **Generate Summary** button.

> While the agent is executed, you will see a spinner icon in the top right corner.

Verify that the AI Agent-generated banking summary and recommendation appear **SIMILAR** to what is shown below.

[REASONING]

Called the extraction tool to parse banking details from the provided document.
Called the validation tool to verify the extracted banking details.

[ANSWER]

Account Verification Status: PASS
Owner Verification Status: PASS
Routing Verification Status: PASS
Account Active Status: FAIL
Compliance Check Status: FAIL
Overall Status: FAIL
Confidence Score: 30%

[RECOMMENDATIONS]

Contact the client to verify and update the banking information.

Do not process any transactions until the account is confirmed active.

Review compliance requirements and ensure KYC/KYB documentation is complete.

Document the validation failure for audit purposes.

Note:

Why SIMILAR and not the same?

Since this is a demo, we instructed the AI Agent to each time provide a different reason for a failing summary. If you try again, you will get different results. For example:

[REASONING]

Called the extraction tool to parse banking details from the provided text.
Called the validation tool to verify the extracted account information.

[ANSWER]

Account Verification Status: FAIL
Owner Verification Status: PASS
Routing Verification Status: PASS
Account Active Status: PASS
Compliance Check Status: FAIL
Overall Status: FAIL
Confidence Score: 30%

[RECOMMENDATIONS]

Contact the client to confirm and correct the bank account details.
Do not process any transactions until verification is successful.
Update the routing number to a valid 9-digit format if necessary.
Document the validation failure for audit purposes.

_67. If you like, repeat steps _63 - _66 to try the second document for Legacy Consulting.

Legacy Consulting - Banking Information.pdf

This customer will always pass the banking test!

[REASONING]

Called the extraction tool to parse banking details from the provided document.
Called the validation tool to verify the extracted account information.

[ANSWER]

Account Verification Status: PASS
Owner Verification Status: PASS
Routing Verification Status: PASS
Account Active Status: PASS
Compliance Check Status: PASS
Overall Status: PASS
Confidence Score: 95%

[RECOMMENDATIONS]

Proceed with setting up the monthly service fee withdrawal as the account is fully verified.
No further action required unless account details change in the future.
Maintain records of this validation for compliance audits.

This concludes the “Use an AI Agent to Validate Banking Information” lab.

Congratulations on completing the lab!